REAL - TIME SELF HEALING NETWORK SYSTEM

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OBJECTIVES

- Continuously monitor the networking to identity faults, anomalies or potential threats in real-time.
- Implement mechanism to automatically fix issues, such as rerouting traffic, restarting services or applying patches.
- Minimize or eliminate network outages by rapidly resolving issues before they impact users or services.

ABSTRACT

The Self-Healing Network System uses XGBoost to autonomously detect and resolve network issues, minimizing downtime. It monitors key parameters and uses machine learning to predict failures and take proactive actions like rerouting traffic. The system enhances cybersecurity by detecting threats and improves reliability, scalability, and security for critical applications like cloud computing and healthcare.

INTRODUCTION

- A self-healing network ensures high availability and reduced downtime by automatically detecting and resolving network issues without manual intervention.
- It uses real-time data analysis and machine learning to identify issues like congestion and device failures, taking corrective actions such as rerouting traffic.
- By automating fault detection and recovery, self-healing networks improve reliability, reduce costs and optimize performance for digital businesses.

PROBLEM IDENTIFICATION

- Network Downtime: Self-healing networks minimize downtime by automatically detecting and fixing failures.
- Manual Intervention: Reduces the need for manual troubleshooting, improving efficiency and reducing errors.
- Scalability Issues: Dynamically adjusts to growing traffic, maintaining performance without manual adjustments.
- Security Vulnerabilities: Automatically detects and mitigates security threats, enhancing network protection.

LITERATURE REVIEW

S No	Title	Author & Year	Journal/ Reference	Remarks
1	Self-Healing in Cloud Networks: AI-Driven Fault Detection and Recovery Strategies for Ubiquitous Connectivity	Arvind Kumar & Kiran L F S, (2021)	IEEE Journal	This paper focuses on improving the self-healing capabilities of cloud networks by using artificial intelligence (AI) to enhance fault detection and recovery
2	AI-Based Self-Healing Mechanism for 5G Networks: Fault Detection and Recovery Techniques	Haibo Wang & Haibin Zhang(2020)	IEEE Journal	Focusing on 5G networks, this paper highlights the need for efficient self-healing mechanisms to support dynamic applications.
3	Self-Healing Networks Using Blockchain and AI for Autonomous Fault Management in SDN-based Networks	Meena B, Yandapally & Nandini M(2023)	IEEE Journal	This paper proposes a hybrid fault management system for software-defined networks (SDN), integrating blockchain and AI to ensure autonomous fault detection and recovery.

LITERATURE REVIEW

S NO	Title	Author & Year	Journal/ Reference	Remarks
4	Self-Healing Networks AI-Based Approaches for Fault Detection and Recovery	Nand Kumar, Elma Sibonghanoy & Groenewald (2023)	IEEE Journal	This focus on machine learning (ML) techniques to automate fault detection and recovery, minimizing downtime and improving network reliability.
5	A Real-Time Fault Recovery System for Smart Grids Using Machine Learning Algorithms	Tarek S,Hegazy S, Ahmed S , Mahmoud M & Elgamal M (2021)	Journals of Internet of Things	The authors present a system that automatically predicts and isolates faults, rerouting power through alternative paths to ensure minimal disruption.
6	Deep Reinforcement Learning for Real-Time Fault Recovery in Cyber-Physical Systems	Yi Xie, Zhihao Sun & Yang Liu (2021)	IEEE International Conference	The result is an adaptive system that can quickly learn to handle new fault conditions, improving the resilience and reliability of CPS in real-time, particularly in complex and dynamic environments.

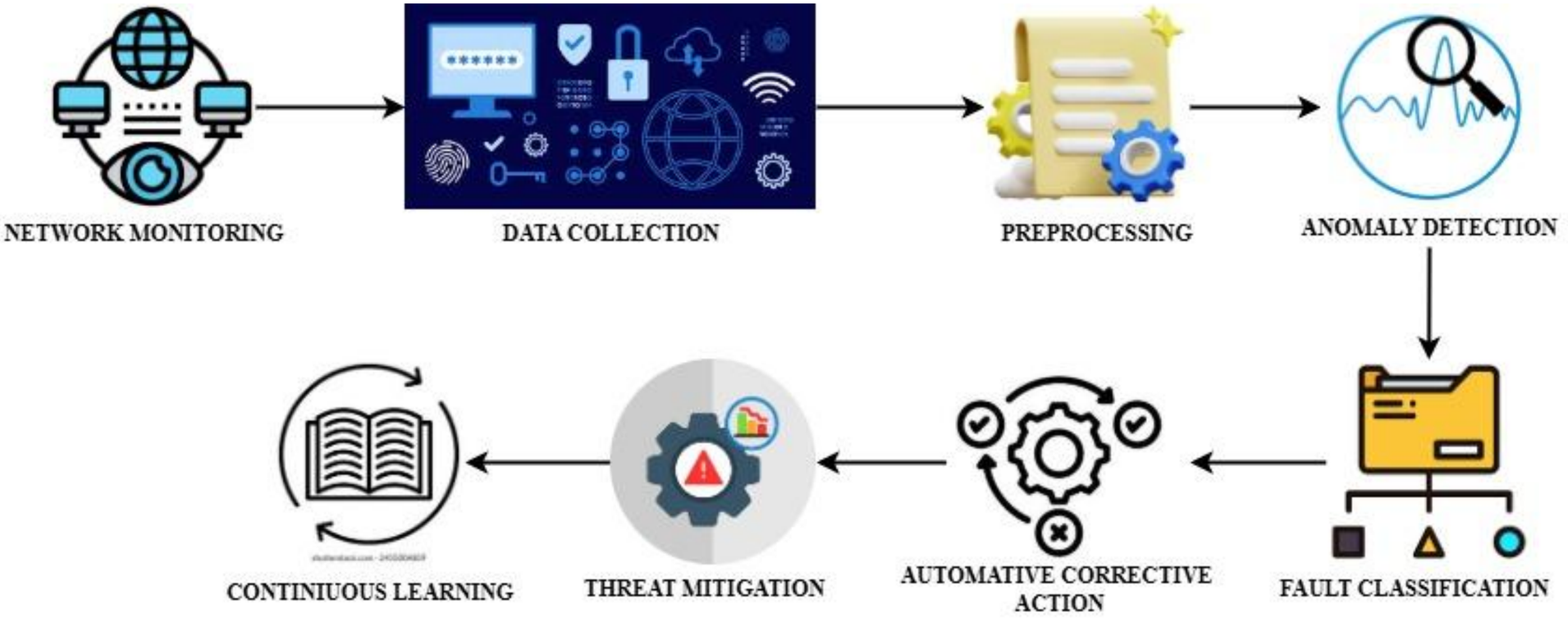
EXISTING SYSTEM

- Manual Monitoring: Traditional networks rely on manual monitoring and static rule-based systems to detect and resolve issues.
- Human Error and Downtime: These methods are time-consuming, error-prone, and lead to prolonged downtime.
- Reactive Fault Detection: Fault detection is reactive, addressing issues only after they affect performance.
- Lack of Predictive Capability: Traditional systems lack predictive features and depend on human intervention and predefined solutions.

PROPOSED SYSTEM

- The system reduces manual intervention by automatically detecting, diagnosing, and resolving network issues while utilizing XGBoost and machine learning for predictive analytics to prevent potential failures.
- Continuous monitoring optimizes network performance and resource allocation, while AI-driven cybersecurity measures detect and mitigate threats like DDoS attacks and unauthorized access.
- Automated corrective actions, such as rerouting traffic or activating backup paths, ensure seamless service and minimize disruptions.

ARCHITECTURE DIAGRAM



SYSTEM REQUIREMENTS

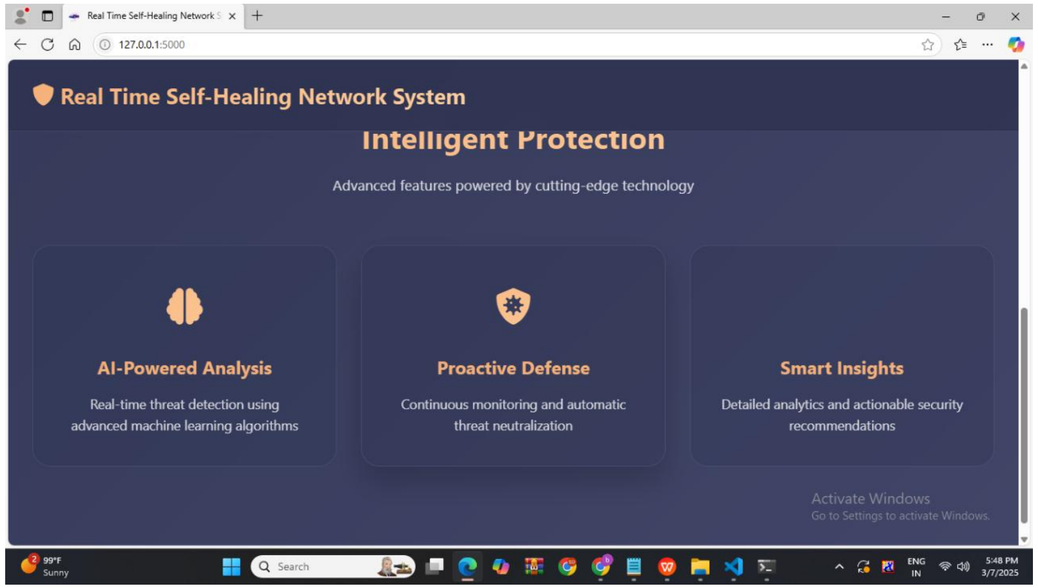
HARDWARE REQUIREMENT:-

PROCESS : INTEL® CORE™ I9-14900K 3.20 GHZ
RAM : 16 GB
HARD DISK : 1 TB

SOFTWARE REQUIREMENT:-

FRONT END	-	HTML,CSS
BACK END	-	PYTHON
FRAMEWORK	-	FLASK

OUTPUT



The screenshot showcases the homepage of a web application titled "Real Time Self-Healing Network System", which is designed to enhance cybersecurity using advanced AI and machine learning technologies. The interface features a sleek, dark-themed layout with bold, peach-colored typography that highlights the key message: "Secure Your Digital Future." Below the headline, a short description explains the system’s core functionality real-time detection of cyber threats and intelligent analysis to ensure proactive protection of digital assets. Prominent action buttons labeled "Start Protection" and "Watch Demo" guide users towards engaging with the system.

ADVANTAGES

- The system automates network management tasks, reducing manual intervention and improving efficiency.
- It identifies issues early and resolves them before they affect performance.
- The system enhances network reliability by quickly addressing problems.
- The system learns from data and optimizes its performance over time.

APPLICATIONS

- Maintains service availability by resolving issues like signal disruptions and hardware failures without manual intervention.
- Supports real-time communication between vehicles and infrastructure, resolving network issues to maintain safety and efficiency.
- Provides reliable and secure communication in harsh environments or during cyber-attacks, ensuring operational continuity.

CONCLUSION

- The system uses XGBoost and machine learning to proactively identify and resolve issues before they affect performance.
- It combines continuous monitoring, predictive analytics and cybersecurity to maintain smooth operation, improve security and reduce the chances of downtime.
- By automating the network management process, the system reduces the need for manual intervention, boosts network performance and increases the network’s overall resilience.

REFERENCES

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THANK YOU